

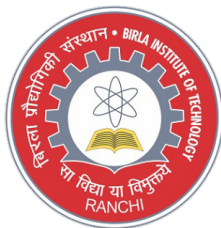
EE24203 Analog and Digital Circuits

Module 3: Field Effect Transistors

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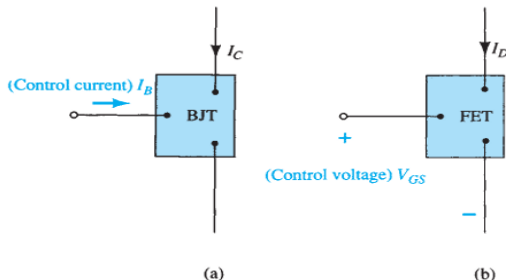
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Junction FET (JFET)

- The field-effect transistor (FET) is a three-terminal device.
- The primary difference between the BJT and JFET is that the BJT transistor is a current-controlled device as depicted in Fig. 1a, while the JFET transistor is a voltage-controlled device as shown in Fig. 1b. In other words, the current I_C in Fig. 1a is a direct function of the level of I_B .
- For the FET the current I will be a function of the voltage V_{GS} applied to the input circuit as shown in Fig. 1b.



Junction FET (JFET)

- Just as there are npn and pnp bipolar transistors, there are n-channel and p-channel field-effect transistors.
- However, it is important to keep in mind that the BJT transistor is a bipolar device—the prefix bi- revealing that the conduction level is a function of two charge carriers, electrons and holes.
- The FET is a unipolar device depending solely on either electron (n-channel) or hole (p-channel) conduction.
- The term field-effect in the chosen name is due to an electric field established by the charges present that will control the conduction path of the output circuit without the need for direct contact between the controlling and controlled quantities.

- Characteristics of JFET:
 - high input impedance
 - FETs are more temperature stable than BJTs
 - FETs are usually smaller in construction than BJTs, making them particularly useful in integrated-circuit (IC) chips.
- Two types of FETs:
 - the junction field-effect transistor (JFET) and
 - the metal-oxide-semiconductor field-effect transistor (MOSFET).
- The MOSFET category is further broken down into:
 - depletion type MOSFET and
 - enhancement type MOSFET

CONSTRUCTION AND CHARACTERISTICS OF N-Channel JFET

- The JFET is a three-terminal device with one terminal capable of controlling the current between the other two.
- The basic construction of the n-channel JFET is shown in Fig. 2.

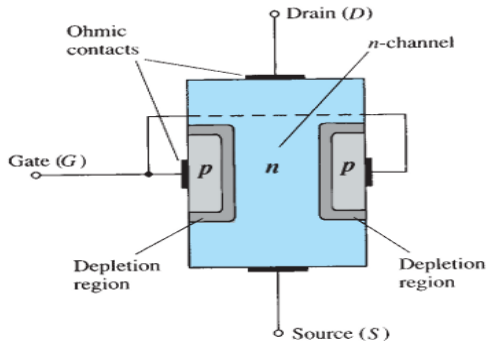


Figure 2: Junction FET

- The major part of the structure is the n-type material, which forms the channel between the embedded layers of p -type material.
- The top of the n-type channel is connected through an ohmic contact to a terminal referred to as the drain (D), whereas the lower end of the same material is connected through an ohmic contact to a terminal referred to as the source (S).
- The two p -type materials are connected together and to the gate (G) terminal.
- The drain and the source are connected to the ends of the n-type channel and the gate to the two layers of p-type material.
- In the absence of any applied potentials the JFET has two p-n junctions under no-bias conditions.
- The result is a depletion region at each junction, as shown in Fig. 2, that resembles the same region of a diode under no-bias conditions. A depletion region is void of free carriers and is therefore unable to support conduction.

$V_{GS} = 0\text{ V}$, $V_{DS} = \text{Some Positive Value}$

- In Fig. 3, a positive voltage V_{DS} is applied across the channel and the gate is connected directly to the source to establish the condition $V_{GS} = 0\text{ V}$.

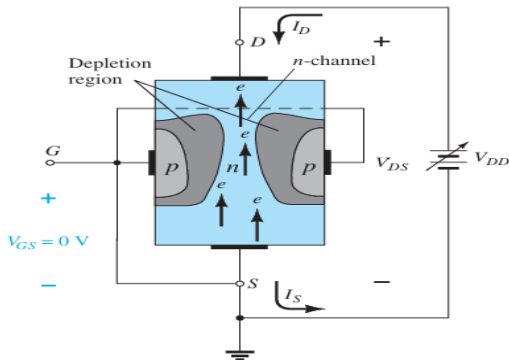


Figure 3: JFET at $V_{GS} = 0\text{ V}$ and $V_{DS} > 0\text{ V}$.

- The result is a gate and a source terminal at the same potential and a depletion region in the low end of each p-material similar to the distribution of the no-bias conditions of Fig. 2.
- The instant the voltage V_{DD} ($=V_{DS}$) is applied, the electrons are drawn to the drain terminal, establishing the conventional current I_D with the defined direction of Fig. 3.
- The path of charge flow clearly reveals that the drain and source currents are equivalent ($I_D = I_S$).
- Under the conditions in Fig. 3, the flow of charge is relatively uninhibited and is limited solely by the resistance of the n-channel between drain and source.

- The depletion region is wider near the top of both p-type materials.
- The reason for the change in width of the region is explained with the help of Fig. 4.

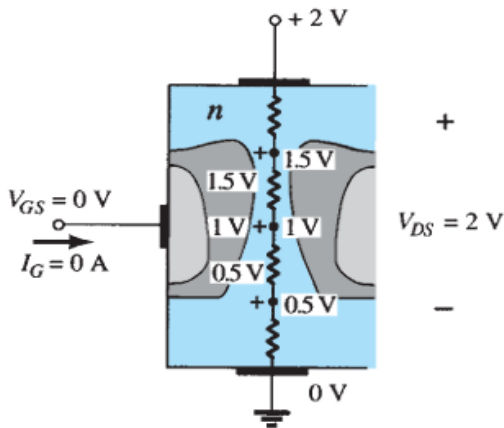


Figure 4: Varying reverse-bias potentials across the p-n junction of an n-channel JFET.

- Assuming a uniform resistance in the n-channel, we can break down the resistance of the channel into the divisions appearing in Fig. 4.
- The current I_D will establish the voltage levels through the channel as indicated on the same figure.
- The result is that the upper region of the p-type material will be reverse-biased by about 1.5 V, with the lower region only reverse-biased by 0.5 V.
- Thus, the greater the applied reverse bias, the wider is the depletion region—hence the distribution of the depletion region as shown in Fig. 4.
- The fact that the p–n junction is reverse-biased for the length of the channel results in a gate current of zero amperes.
- Thus, $I_G = 0$ A is an important characteristic of the JFET.

- As the voltage V_{DS} is increased from 0 V to a few volts, the current will increase as determined by Ohm's law and the plot of I_D versus V_{DS} will appear as shown in Fig. 5.

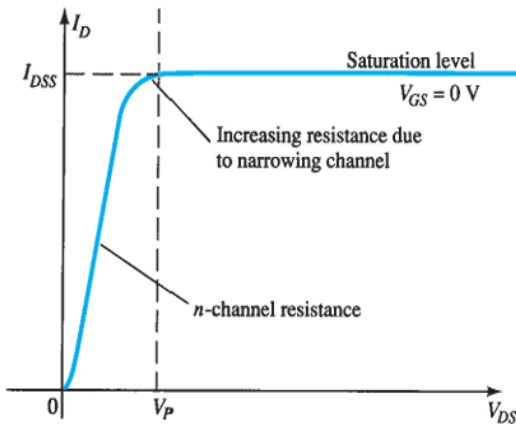


Figure 5: I_D versus V_{DS} for $V_{GS} = 0$ V.

- The relative straightness of the plot reveals that for the region of low values of V_{DS} , the resistance is essentially constant.
- As V_{DS} increases and approaches a level referred to as V_P in Fig. 5, the depletion regions of Fig. 3 will widen, causing a noticeable reduction in the channel width.
- The reduced path of conduction causes the resistance to increase and the curve in the graph of Fig. 5 to occur.
- The more horizontal the curve, the higher the resistance, suggesting that the resistance is approaching "infinite" ohms in the horizontal region.

- If V_{DS} is increased to a level where it appears that the two depletion regions would "touch" as shown in Fig. 6, a condition referred to as pinch-off will result.

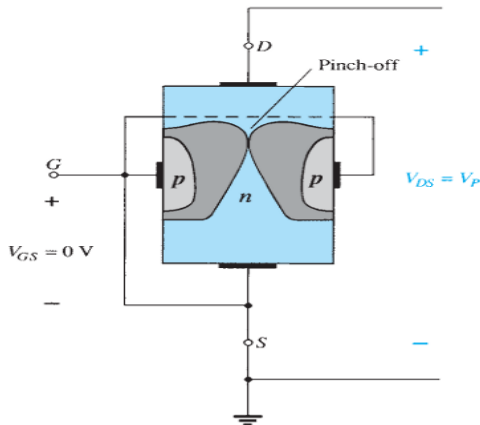


Figure 6: Pinch-off ($V_{GS} = 0 \text{ V}$, $V_{DS} = V_P$).

- The level of V_{DS} that establishes this condition is referred to as the pinch-off voltage and is denoted by V_P , as shown in Fig. 5.
- In actuality, the term pinch-off is a misnomer in that it suggests the current I_D is pinched off and drops to 0 A.
- As shown in Fig. 5, however, this is hardly the case- I_D maintains a saturation level defined as I_{DSS} in Fig. 5.
- In reality a very small channel still exists, with a current of very high density.
- The fact that I_D does not drop off at pinch-off and maintains the saturation level is verified by the following fact:
 - The absence of a drain current would remove the possibility of different potential levels through the n-channel material to establish the varying levels of reverse bias along the p-n junction.
- The result would be a loss of the depletion region distribution that caused pinch-off in the first place.

- As V_{DS} is increased beyond V_P , the region of close encounter between the two depletion regions increases in length along the channel, but the level of I_D remains essentially the same.
- In essence, therefore, once $V_{DS} > V_P$ the JFET has the characteristics of a current source.

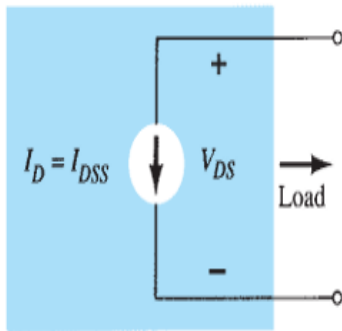
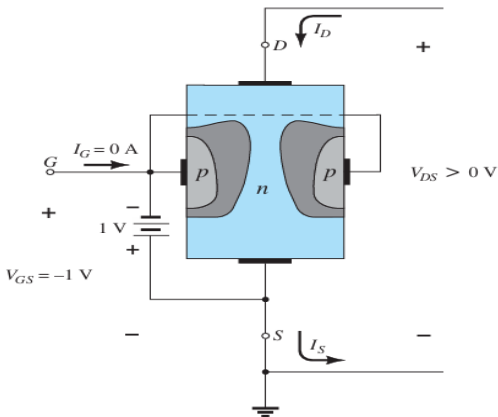


Figure 7

- As shown in Fig. 7, the current is fixed at $I_D = I_{DSS}$, but the voltage V_{DS} (for levels $> V_P$) is determined by the applied load.
- The choice of notation I_{DSS} is derived from the fact that it is the drain-to-source current with a short-circuit connection from gate to source.
- As we continue to investigate the characteristics of the device we will find that:
 - **I_{DSS} is the maximum drain current for a JFET and is defined by the conditions $V_{GS} = 0 \text{ V}$ and $V_{DS} > |V_P|$.**

$$V_{GS} < 0 \text{ V}$$

- The voltage from gate to source, denoted V_{GS} , is the controlling voltage of the JFET.
- For the n-channel device the controlling voltage V_{GS} is made more and more negative from its $V_{GS} = 0 \text{ V}$ level.



- In Fig. 8 a negative voltage of 1 V is applied between the gate and source terminals for a low level of V_{DS} .
- The effect of the applied negative-bias V_{GS} is to establish depletion regions at lower levels of V_{DS} , thus to reach the saturation level at a lower level of V_{DS} ,

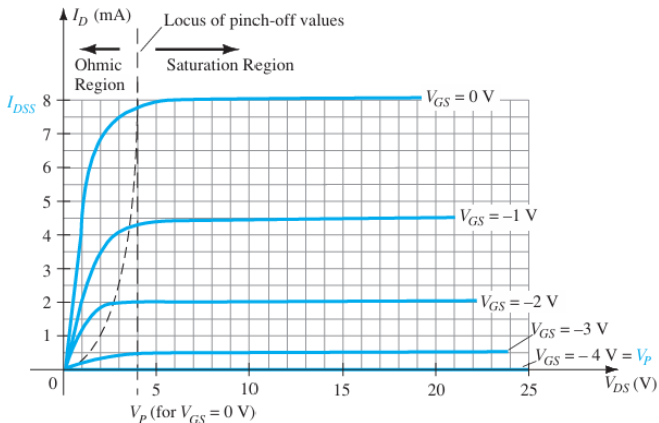


Figure 9: n-Channel JFET characteristics with $I_{DSS} = 8$ mA and $V_P = -4$ V.

- The resulting saturation level for I_D has been reduced and in fact will continue to decrease as V_{GS} is made more and more negative.
- The pinch-off voltage continues to drop in a parabolic manner as V_{GS} becomes more and more negative.
- Eventually, V_{GS} when $V_{GS} = -V_P$ will be sufficiently negative to establish a saturation level that is essentially 0 mA, and for all practical purposes the device has been "turned off".
- In summary: The level of V_{GS} that results in $I_D = 0$ mA is defined by $V_{GS} = V_P$, with V_P being a negative voltage for n-channel devices and a positive voltage for p-channel JFETs.

Voltage Controlled Resistor

- The region to the left of the pinch-off locus of Fig. 9 is referred to as the ohmic or voltage-controlled resistance region.
- In this region the JFET can actually be employed as a variable resistor whose resistance is controlled by the applied gate-to-source voltage.
- The slope of each curve and therefore the resistance of the device between drain and source for $V_{DS} < V_P$ are a function of the applied voltage V_{GS} .
- As V_{GS} becomes more and more negative, the slope of each curve becomes more and more horizontal, corresponding to an increasing resistance level.
- The following equation provides a good first approximation to the resistance level in terms of the applied voltage V_{GS} :

$$r_d = \frac{r_0}{(1 - V_{GS}/V_P)^2} \quad (1)$$

where r_0 is the resistance with $V_{GS} = 0$ V and r_d is the resistance at a particular level of V_{GS} .

p-channel devices

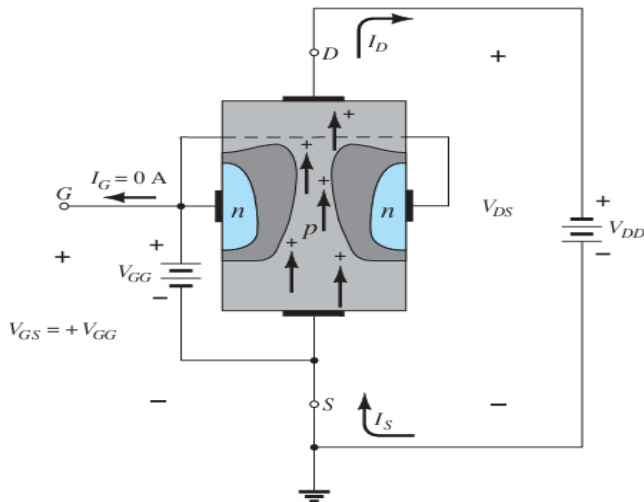


Figure 10: p-Channel JFET

I-V Characteristics of p-channel JFET

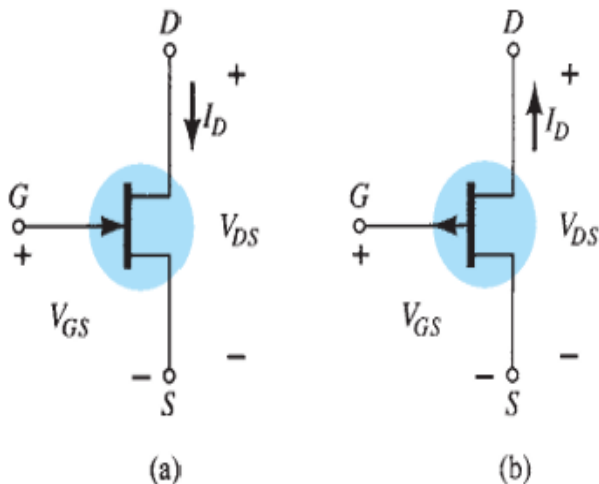


Figure 11: p-Channel JFET characteristics with $I_{DSS} = 6\text{mA}$ and $V_P = 6\text{V}$.

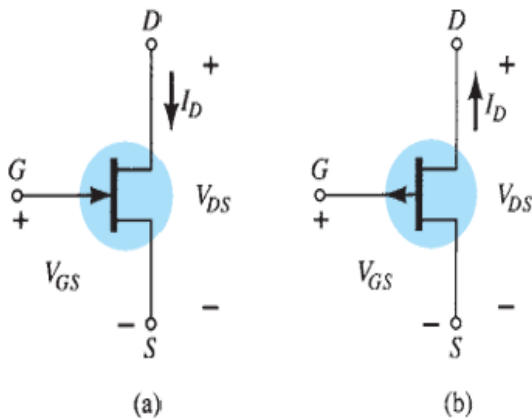
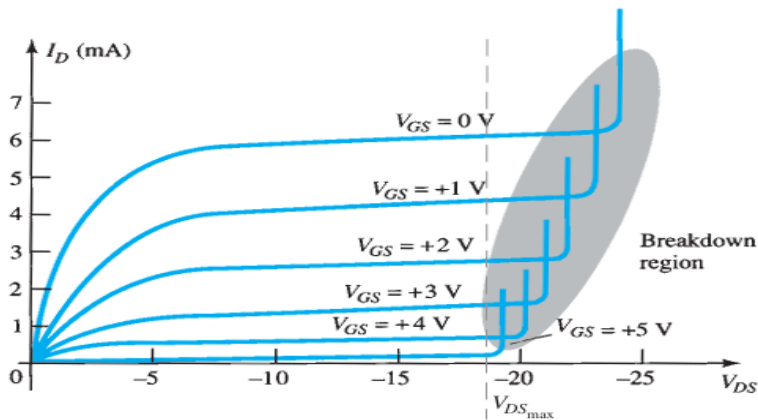


Figure 12: JFET symbols: (a) n-channel; (b) p-channel.

Transfer Characteristics

- The relationship between I_D and V_{GS} is defined by Shockley's equation :

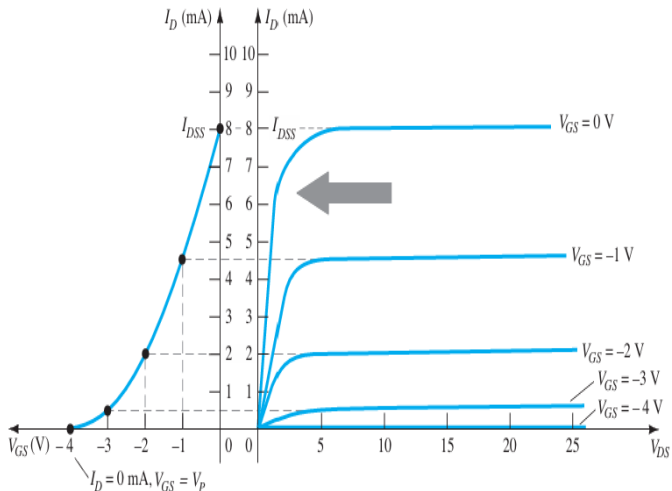
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 \quad (2)$$



- MOSFETs are further broken down into depletion type and enhancement type.
- The terms depletion and enhancement define their basic mode of operation; the name MOSFET stands for metal–oxide–semiconductor field-effect transistor.

Depletion Type MOSFET

- The basic construction of the n-channel depletion-type MOSFET is provided in Fig. 14.



- A slab of p-type material is formed from a silicon base and is referred to as the substrate.
- In some cases the substrate is internally connected to the source terminal.
- However, many discrete devices provide an additional terminal labeled SS, resulting in a four-terminal device.
- The source and drain terminals are connected through metallic contacts to n-doped regions linked by an n-channel as shown in the figure.
- The gate is also connected to a metal contact surface but remains insulated from the n-channel by a very thin silicon dioxide (SiO_2) layer.
- SiO_2 is a type of insulator referred to as a dielectric, which sets up opposing electric fields within the dielectric when exposed to an externally applied field.

- The fact that the SiO_2 layer is an insulating layer means that:
 - **There is no direct electrical connection between the gate terminal and the channel of a MOSFET.**
- It is the insulating layer of SiO_2 in the MOSFET construction that accounts for the very desirable high input impedance of the device.
- The reason for the label metal–oxide–semiconductor FET is: metal for the drain, source, and gate connections; oxide for the silicon dioxide insulating layer; and semiconductor for the basic structure on which the n- and p-type regions are diffused.
- The insulating layer between the gate and the channel has resulted in another name for the device: insulated-gate FET, or IGFET.

Basic Operation and Characteristics

- In Fig. 15 the gate-to-source voltage is set to 0 V by the direct connection from one terminal to the other, and a voltage V_{DD} is applied across the drain-to-source terminals.

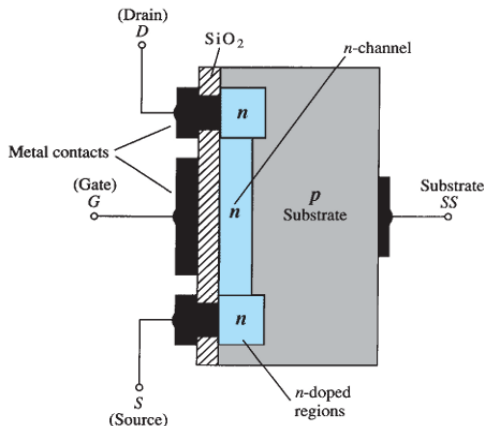


Figure 15: n-Channel depletion-type MOSFET with $V_{GS} = 0$ V and applied

- The result is an attraction of the free electrons of the n-channel for the positive voltage at the drain.
- The result is a current similar to that flowing in the channel of the JFET. In fact, the resulting current with $V_{GS} = 0$ V continues to be labeled I_{DSS} .

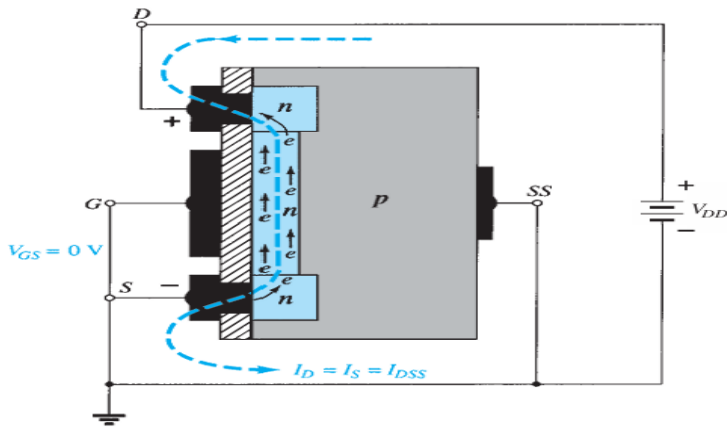


Figure 16: Drain and transfer characteristics for an n-channel depletion-type MOSFET

- In Fig. 16, V_{GS} is set at a negative voltage such as -1 V. The negative potential at the gate will tend to pressure electrons toward the p-type substrate (like charges repel) and attract holes from the p-type substrate (opposite charges attract) as shown in Fig. . Depending on the and holes will occur that will reduce the number of free electrons in the n -channel available for conduction.
- The more negative the bias, the higher is the rate of recombination.
- The resulting level of drain current is therefore reduced with increasing negative bias for V_{GS} , as shown in Fig. for $V_{GS} = -1$ V, -2 V, and so on, to the pinch-off level of -6 V.

CS biasing of JFET and MOSFET

- The biasing levels for a silicon transistor configuration can be obtained using the characteristic equations $V_{BE} = 0.7 \text{ V}$, $I_C = \beta I_B$, and $I_C = \alpha I_E$.
- The linkage between input and output variables is provided by β , which is assumed to be fixed in magnitude for the analysis to be performed.
- The fact that beta is a constant establishes a linear relationship between I_C and I_B .
- Doubling the value of I_B will double the level of I_C , and so on.
- For the field-effect transistor, the relationship between input and output quantities is nonlinear due to the squared term in Shockley's equation.
- Linear relationships result in straight lines when plotted on a graph of one variable versus the other, while nonlinear functions result in curves as obtained for the transfer characteristics of a JFET.

- The nonlinear relationship between I_D and V_{GS} can complicate the mathematical approach to the dc analysis of FET configurations.
- A graphical approach may limit solutions to tenths-place accuracy, but it is a quicker method for most FET amplifiers.
- Since the graphical approach is in general the most popular, the analysis of this chapter will have a graphical orientation rather than direct mathematical techniques.
- Another distinct difference between the analysis of BJT and FET transistors is that the input controlling variable for a BJT transistor is a current level, while for the FET a voltage is the controlling variable.
- In both cases, however, the controlled variable on the output side is a current level that also defines the important voltage levels of the output circuit.

END OF MODULE 2